Angelo Reyes

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Education

Texas A&M University — Expected May 2029
B.S. Electrical Engineering · GPA 3.66 — *College Station, TX*

Experience

Controls & Data Acquisition Engineer — Mar 2026 – Present
Texas A&M Baja SAE 2027 — Controls / DAQ Subteam — *College Station, TX*

- Own sensor selection, signal conditioning, firmware, and PCB design for a **14-sensor, 12 V distributed CAN network** on a single-seat off-road race car competing against ~100 university teams.
- Classified every acquisition channel by **signal type** to drive front-end design: an inductive RPM pickup returns **analog AC** whose amplitude scales with shaft speed, a Hall effect sensor (GS100502) returns an **already-digital open-collector** square wave, and a 1000 kg load cell returns **analog differential** mV/V — three channels needing three different circuits.
- Built dynamometer test firmware on ESP32-C3 — interrupt-driven period capture with `IRAM_ATTR` handlers, HX711 24-bit load cell with EMA filtering, hardware LEDC PWM — with **fail-safe detection commanding neutral on stale sensor data** on a rig capable of over 3,600 lbf.
- Specified harness and connector architecture to an **IP67** standard: Deutsch DT for sealed, positively-latched, field-serviceable runs; Molex Micro-Fit 3.0 board-mount pigtails with pegs transferring vibration load into the board rather than solder joints. Produced pinout and wire-gauge mapping.
- Drove **scope reduction** after design review: re-sorted the sensor list with Powertrain, cutting instrumentation nobody would consume and adding a time-of-flight sheave sensor that closed a real gap — **CVT belt slip**, previously indistinguishable from a legitimate ratio change.

Hardware Security Research Assistant — Nov 2025 – Jun 2026
Texas A&M University — *College Station, TX*

- Tuned CPU architecture for zero-knowledge proof protocols across NTT, MSM, hash, and sum-check algorithms alongside FPGA/VLSI design; implemented NIST FIPS post-quantum lattice cryptography with learning-with-errors key encapsulation.

Technical Lead — ThinkTank TAMU — Aug 2025 – May 2026
AIAA Undergraduate Space Mission Design Challenge — *College Station, TX*

- Designed CubeSat telemetry, solar weather monitoring, and communication subsystems, including heliocentric orbits holding a link across black-shadow periods in the Mars–Earth line; led concept of operations and trade studies.

Baja SAE Engineering Projects

Engine Dynamometer Brake Actuator — 2026
Trade study · force transmission analysis · ANSYS FEA · ESP32-C3 · PID

- Selected a slab ball-screw architecture over three alternatives on a **control** argument, not a mechanical one: a lever arm's mechanical advantage varies with stroke position, so a single set of PID gains cannot be correct across its range. Linear actuation makes the plant behave identically everywhere.
- Sized the transmission to the **controller's** 40 A continuous limit rather than the motor's 133 A stall — the real operating point is set by whichever component gives out first. Delivered **16,342 N (3,674 lbf)** at a 20:1 reduction through an SFU1204 4 mm-lead screw, **3.7**× the 1,000 lbf operational target and inside the screw's 5,000 lbf axial rating.
- Caught a lead-screw error that had propagated silently through the entire force chain: a 4 mm lead recorded as 5 mm understated every force by **25%** and overstated linear speed by **20%**, since lead sits in the denominator of the governing equation. Re-derived and re-typeset the analysis.
- Ran ANSYS Static Structural FEA on the load-bearing push slab and reported the resulting **FOS of 1.02** openly at design review — a stress concentration at a narrow waist rather than a material inadequacy — revising the geometry toward a 1.25 target.

Rear CAN Node PCB — 2026
KiCad · STM32G431RBTx · FDCAN · signal integrity · transient protection

- Designed a complete schematic through ERC for a distributed CAN node acquiring engine RPM, CVT secondary RPM, and rear shock potentiometers; selected the STM32G431RBTx against a weighted decision matrix for native FDCAN, hardware timer capture removing ISR jitter, and an FMAC accelerator offloading analog filtering.
- **Rebuilt the inductive RPM front end after bench testing invalidated the design.** A 147 kΩ divider drawing only 82 µA still collapsed the sensor to 2.4 V, below the logic threshold. Replaced attenuation with a **2N3904 used as a switch**, regenerating logic level from board supply on ~0.38 mA of base current so the sensor's 12 V never reaches the MCU.
- Engineered power and analog signal integrity: a **600 Ω at 100 MHz ferrite** isolating VDDA — chosen over an inductor, which would resonate with the decoupling network — parallel 1 µF / 10 nF bypass covering bulk and low-ESL response, and per-supply-pin 100 nF sized against bond-wire inductance.
- Specified transient protection on every externally exposed pin — TVS diodes across all power rails, both RPM inputs, and the CAN pair — plus **jumper-selectable 120 Ω bus termination**, since which node sits at a physical end of the bus changes with vehicle wiring.

Additional Projects

- **Liquid-Cooled Robotic Manipulator** (2025–Present) — Independent research build attacking the thermal ceiling on actuator torque density. Designed and manufactured a PCB implementing **Field-Oriented Control** of frameless BLDC phase windings, with a soft limiter tapering commanded current against a live winding-temperature estimate rather than tripping a hard cutoff.
- **Sensor and instrumentation builds** — PCB for a capacitance-based plant health monitor (AD9833 sweep, FDC2214/MPR121 capacitive front ends, AS7263 NIR cross-check; three-person team); an ORP salinity instrument built as a galvanic cell from scratch (Ag/AgCl reference in KCl vs. inert platinum, hand-cast agar salt bridge, **250–5000 ppm**); and an iOS pixel grid bit-packed at **one bit per pixel** to fit BLE's ~20-byte payload, an 8× reduction.

Technical Skills

Hardware & PCB: KiCad capture and ERC, decoupling and power distribution strategy, ferrite analog supply isolation, TVS transient protection, open-collector and level-shifted sensor front ends, CAN transceiver and bus termination, STM32 pin planning, Deutsch DT / Molex Micro-Fit harness design to IP67
Embedded & Controls: C++ bare-metal and Arduino, ISRs with IRAM placement, hardware timer input capture, LEDC PWM, 24-bit ADC interfacing, digital filtering, fault detection, CAN / FDCAN, PID with feedforward, Field-Oriented Control, latency budgeting, fail-safe state design
Analysis & Tools: ANSYS Static Structural FEA, force transmission modeling, weighted decision matrices, DAQ methodology, test plan development · C++, Python, Verilog, Swift, Git/GitHub, LaTeX, STM32CubeIDE, Onshape, SolidWorks